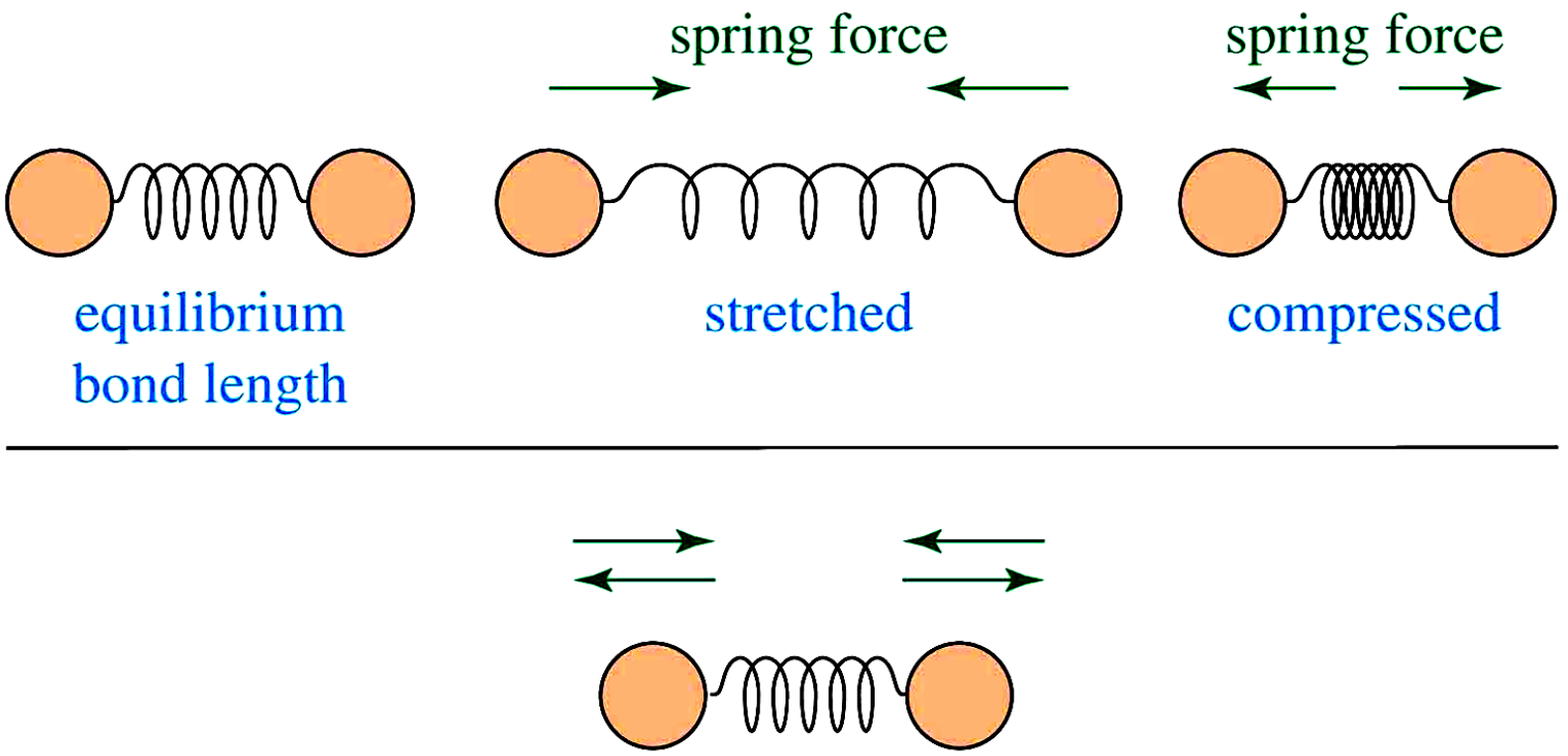
Infrared radiation is largely thermal energy.
It induces stronger **molecular vibrations** in covalent bonds, which can be viewed as springs holding together two masses, or atoms.

Specific bonds respond to (absorb) **specific** frequencies



3.1 THE VIBRATING DIATOMIC MOLECULE

3.1.1 The Energy of a Diatomic Molecule

When two atoms combine to form a stable covalent molecule (e.g., HCl gas) they may be said to do so because of some internal electronic rearrangement. This is not the place to discuss the detailed mechanisms of chemical bond formation; we may simply look on the phenomenon as a balancing of forces. On the one hand there is a repulsion between the positively charged nuclei of both atoms, and between the negative electron 'clouds'; on the other there is an attraction between the nucleus of one atom and the electrons of the other, and vice versa. The two atoms settle at a mean internuclear distance such that these forces are just balanced and the energy of the whole system is at a minimum. Attempt to squeeze the atoms more closely together and the repulsive force rises rapidly; attempt to pull them further apart and we are resisted by the attractive force. In either case an attempt to distort the bond requires an input of energy and so we may plot energy against internuclear distance as in Fig. 3.1. At the minimum the internuclear distance is referred to as the equilbrium distance $r_{eq.}$, or more simply, as the bond length.

The compression and extension of a bond may be likened to the behaviour of a spring and we may extend the analogy by assuming that the bond, like a spring, obeys Hooke's law. We may then write

$$f = -k(r - r_{eq.}) \tag{3.1}$$

where f is the restoring force, k the force constant, and r the internuclear distance. In this case the energy curve is parabolic and has the form

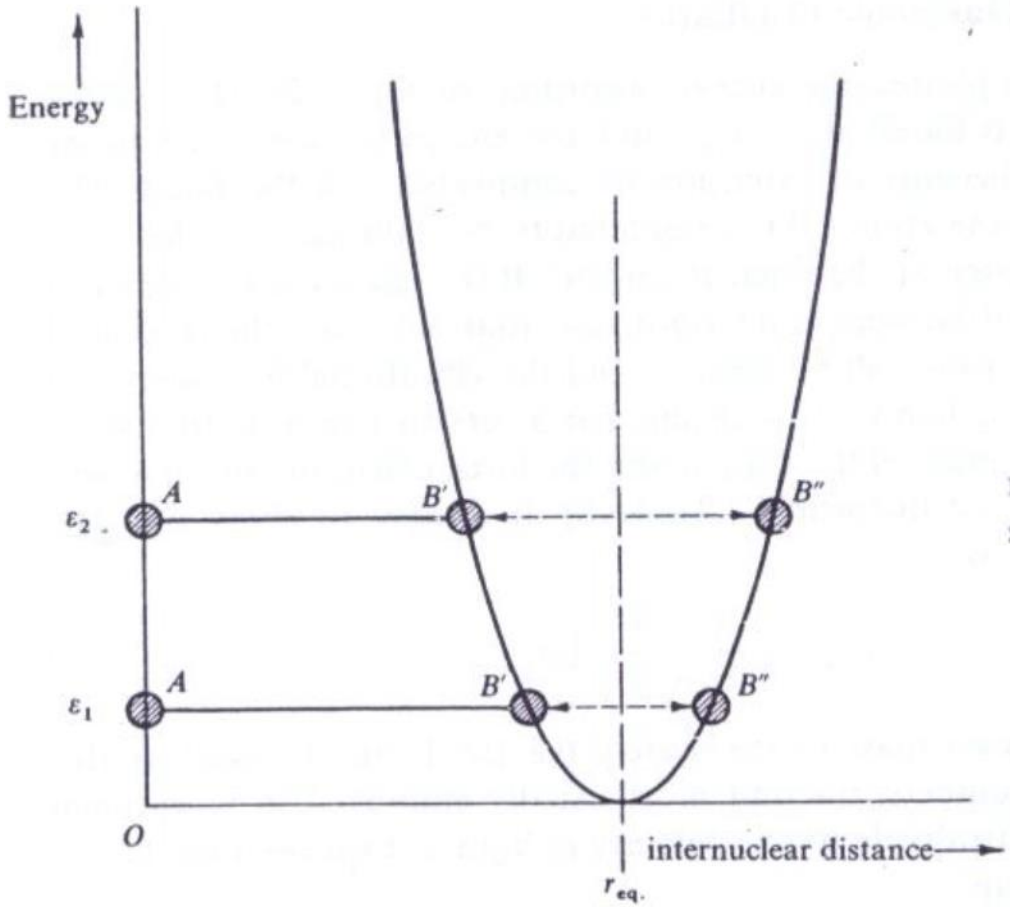


Figure 3.1 Parabolic curve of energy plotted against the extension or compression of a spring obeying Hooke's law.

$$E = \tfrac{1}{2}k(r - r_{eq.})^2 \tag{3.2}$$

This model of a vibrating diatomic molecule—the so-called simple harmonic oscillator model—while only an approximation, forms an excellent starting point for the discussion of vibrational spectra.

3.1.2 The Simple Harmonic Oscillator

In Fig. 3.1 we have plotted the energy according to Eq. (3.2). The zero of curve and equation is found at $r = r_{eq.}$, and any energy in excess of this, for example, ε_1, arises because of extension or compression of the bond. The figure shows that if one atom (A) is considered to be stationary on the $r = 0$ axis, the other will oscillate between B' and B''. If the energy is increased to ε_2 the oscillation will become more vigorous—that is to say, the degree of compression or extension will be greater—but the vibrational frequency will not change. An elastic bond, like a spring, has a certain vibration frequency dependent upon the mass of the system and the force constant, but independent of the amount of distortion. Classically it is easy to show that the oscillation frequency is:

$$\omega_{osc.} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}} \quad \text{Hz} \tag{3.3}$$

where μ is the reduced mass of the system (cf. Eq. (2.9)). To convert this frequency to wavenumbers, the unit most usually employed in vibrational spectroscopy, we must divide by the velocity of light, c, expressed in cm s^{-1} (cf. Sec. 1.1), obtaining:

$$\bar{\omega}_{osc.} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \quad \text{cm}^{-1} \tag{3.4}$$

Vibrational energies, like all other molecular energies, are quantized, and the allowed vibrational energies for any particular system may be calculated from the Schrödinger equation. For the simple harmonic oscillator these turn out to be:

$$E_v = (v + \tfrac{1}{2})h\omega_{\text{osc.}} \quad \text{joules} \qquad (v = 0, 1, 2, \ldots) \tag{3.5}$$

where v is called the *vibrational quantum number*. Converting to the spectroscopic units, cm^{-1}, we have:

$$\varepsilon_v = \frac{E_v}{hc} = (v + \tfrac{1}{2})\bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \tag{3.6}$$

as the only energies allowed to a simple harmonic vibrator. Some of these are shown in Fig. 3.2.

In particular we should notice that the *lowest* vibrational energy, obtained by putting $v = 0$ in Eq. (3.5) or (3.6), is

$$E_0 = \tfrac{1}{2}h\omega_{\text{osc.}} \quad \text{joules} \qquad [\omega_{\text{osc.}} \text{ in Hz}]$$

or

$$\varepsilon_0 = \tfrac{1}{2}\bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \qquad [\bar{\omega}_{\text{osc.}} \text{ in cm}^{-1}] \tag{3.7}$$

The implication is that the diatomic molecule (and, indeed, *any* molecule) can never have zero vibrational energy; the atoms can never be completely at rest relative to each other. The quantity $\frac{1}{2}h\omega_{\text{osc.}}$ joules or $\frac{1}{2}\bar{\omega}_{\text{osc.}}$ cm^{-1} is known as the zero-point energy; it depends only on the classical vibration frequency and hence (Eq. (3.3) or (3.4)) on the strength of the chemical bond and the atomic masses.

The prediction of zero-point energy is the basic difference between the wave mechanical and classical approaches to molecular vibrations. Classical mechanics could find no objection to a molecule possessing no vibrational energy but wave mechanics insists that it must always vibrate to some extent; the latter conclusion has been amply borne out by experiment.

Further use of the Schrödinger equation leads to the simple *selection*

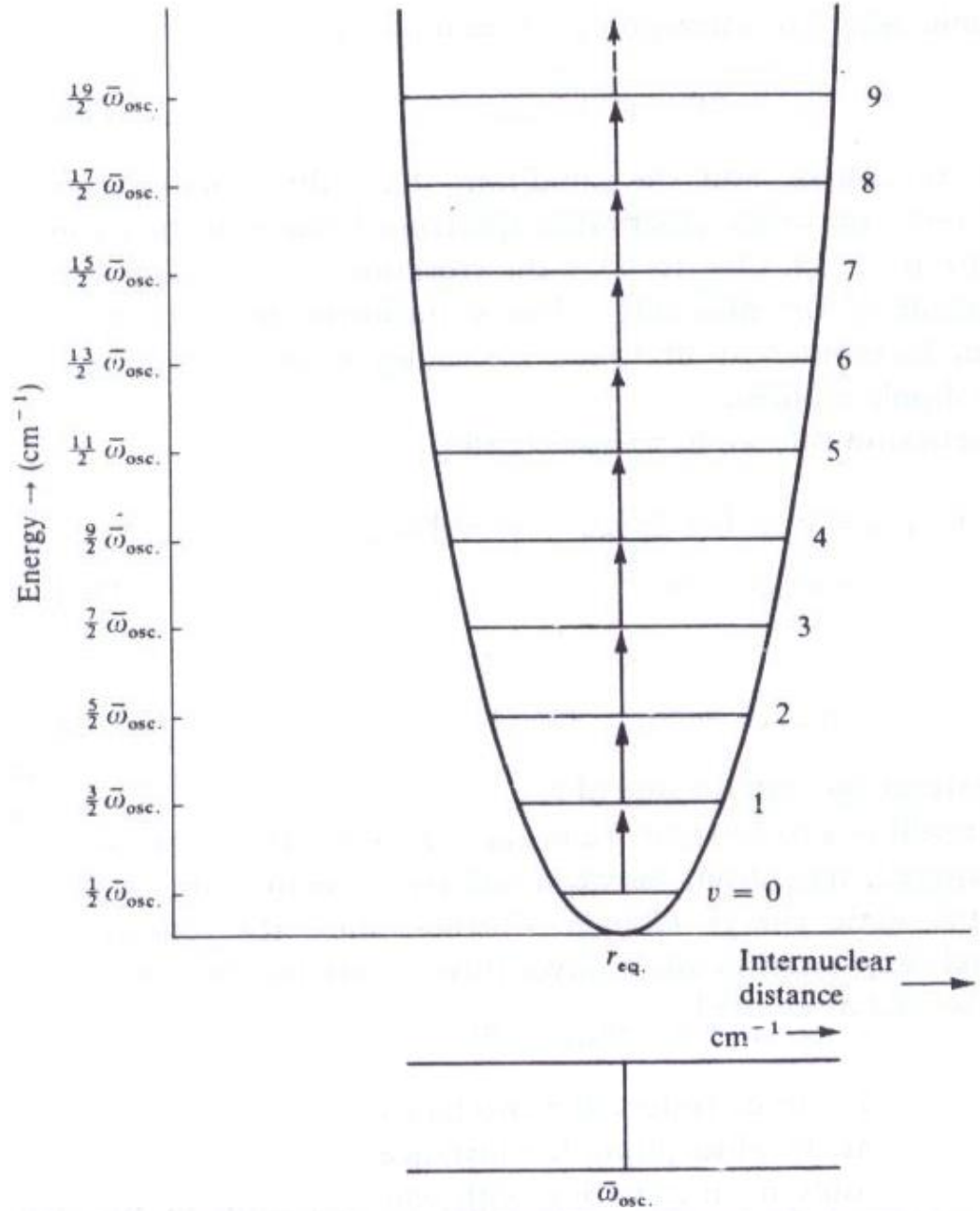


Figure 3.2 The allowed vibrational energy levels and transitions between them for a diatomic molecule undergoing simple harmonic motion.

rule for the harmonic oscillator undergoing vibrational changes:

$$\Delta v = \pm 1 \tag{3.8}$$

To this we must, of course, add the condition that vibrational energy changes will only give rise to an observable spectrum if the vibration can interact with radiation, i.e. (cf. Chapter 1), if the vibration involves a change in the dipole moment of the molecule. Thus vibrational spectra will be observable only in heteronuclear diatomic molecules since homonuclear molecules have no dipole moment.

Applying the selection rule we have immediately:

$$\begin{aligned} \varepsilon_{v+1 \to v} &= (v + 1 + \tfrac{1}{2})\bar{\omega}_{\text{osc.}} - (v + \tfrac{1}{2})\bar{\omega}_{\text{osc.}} \\ &= \bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \end{aligned} \tag{3.9a}$$

for emission and

$$\varepsilon_{v \to v+1} = \bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \tag{3.9b}$$

for absorption, whatever the initial value of v.

Such a simple result is also obvious from Fig. 3.2—since the vibrational levels are equally spaced, transitions between any two neighbouring states will give rise to the same energy change. Further, since the difference between energy levels expressed in cm^{-1} gives directly the wavenumber of the spectral line absorbed or emitted

$$\bar{v}_{\text{spectroscopic}} = \varepsilon = \bar{\omega}_{\text{osc.}} \quad \text{cm}^{-1} \tag{3.10}$$

This, again, is obvious if one considers the mechanism of absorption or emission in classical terms. In absorption, for instance, the vibrating molecule will absorb energy only from radiation with which it can coherently interact (cf. Fig. 1.8) and this must be radiation of its own oscillation frequency.

3.1.3 The Anharmonic Oscillator

Real molecules do not obey exactly the laws of simple harmonic motion; real bonds, although elastic, are not so homogeneous as to obey Hooke's law. If the bond between atoms is stretched, for instance, there comes a point at which it will break—the molecule dissociates into atoms. Thus although for small compressions and extensions the bond may be taken as perfectly elastic, for larger amplitudes—say greater than 10 per cent of the bond length—a much more complicated behaviour must be assumed. Figure 3.3 shows, diagrammatically, the shape of the energy curve for a typical diatomic molecule, together with (dashed) the ideal, simple harmonic parabola.

A purely empirical expression which fits this curve to a good approximation was derived by P. M. Morse, and is called the Morse function:

$$E = D_{\text{eq.}}[1 - \exp\{a(r_{\text{eq.}} - r)\}]^2 \tag{3.11}$$

where a is a constant for a particular molecule and $D_{\text{eq.}}$ is the dissociation energy.

When Eq. (3.11) is used instead of Eq. (3.2) in the Schrödinger equation, the pattern of the allowed vibrational energy levels is found to be:

$$\varepsilon_v = (v + \tfrac{1}{2})\bar{\omega}_e - (v + \tfrac{1}{2})^2\bar{\omega}_e x_e \quad \text{cm}^{-1} \qquad (v = 0, 1, 2, \ldots) \tag{3.12}$$

where $\bar{\omega}_e$ is an oscillation frequency (expressed in wavenumbers) which we shall define more closely below, and x_e is the corresponding anharmonicity constant which, for bond stretching vibrations, is always small and positive

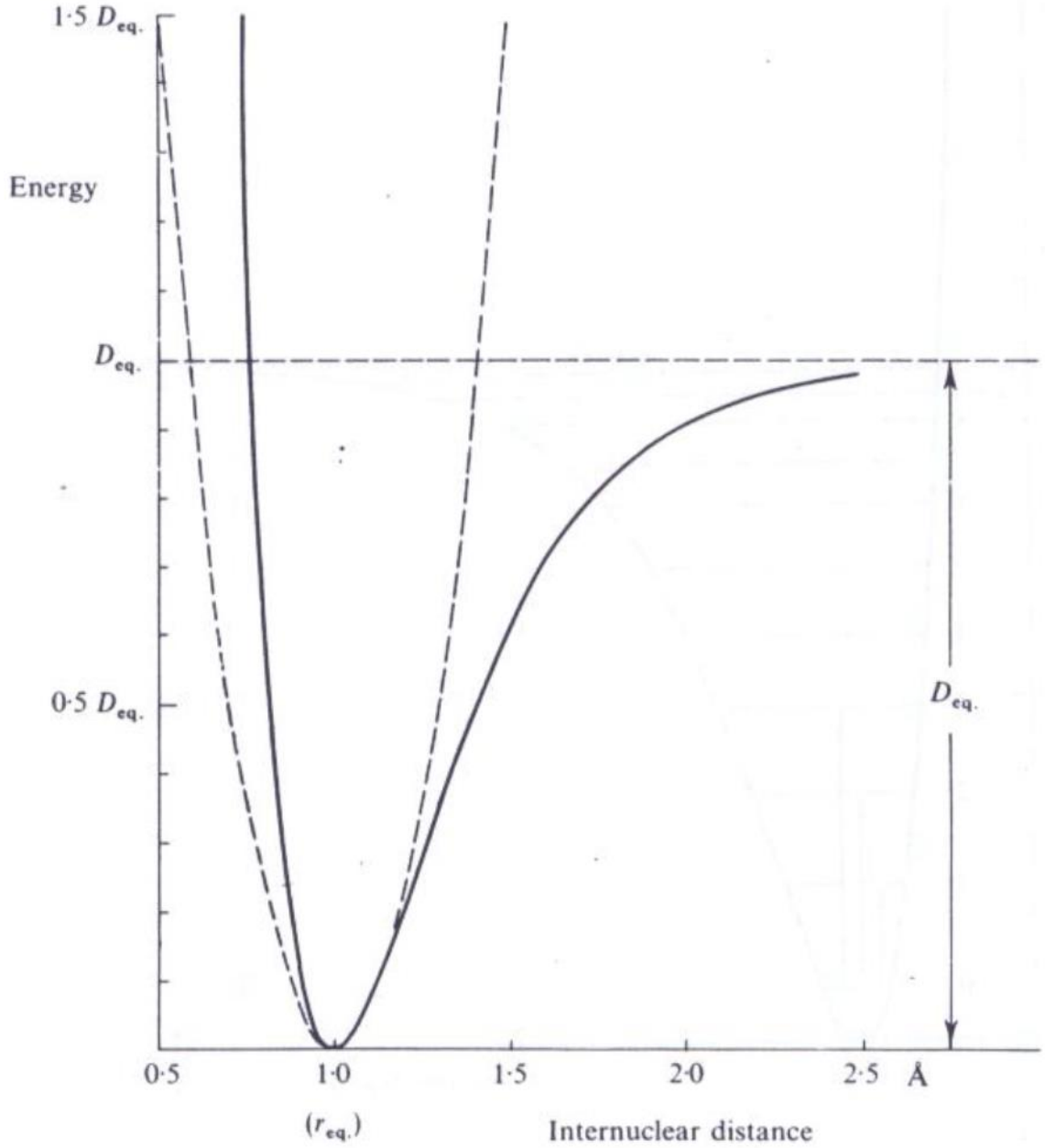


Figure 3.3 The Morse curve: the energy of a diatomic molecule undergoing anharmonic extensions and compressions.

($\approx +0{\cdot}01$), so that the vibrational levels crowd more closely together with increasing v. Some of these levels are sketched in Fig. 3.4.

It should be mentioned that Eq. (3.12), like (3.11), is an approximation only; more precise expressions for the energy levels require cubic, quartic, etc., terms in $(v + \frac{1}{2})$ with anharmonicity constants y_e, z_e, etc., rapidly diminishing in magnitude. These terms are important only at large values of v, and we shall ignore them.

If we rewrite Eq. (3.12), for the anharmonic oscillator, as:

$$\varepsilon_v = \bar{\omega}_e \{1 - x_e(v + \tfrac{1}{2})\}(v + \tfrac{1}{2}) \tag{3.13}$$

and compare with the energy levels of the *harmonic* oscillator (Eq. (3.6)), we see that we can write:

$$\bar{\omega}_{\text{osc.}} = \bar{\omega}_e \{1 - x_e(v + \tfrac{1}{2})\} \tag{3.14}$$

Thus the anharmonic oscillator behaves like the harmonic oscillator but with an oscillation frequency which decreases steadily with increasing v. If we now consider the hypothetical energy state obtained by putting $v = -\frac{1}{2}$ (at which, according to Eq. (3.13), $\varepsilon = 0$) the molecule would be at the equilibrium point with zero vibrational energy. Its oscillation frequency (in cm^{-1}) would be:

$$\bar{\omega}_{\text{osc.}} = \bar{\omega}_e$$

Thus we see that $\bar{\omega}_e$ may be defined as the (hypothetical) *equilibrium oscillation frequency* of the anharmonic system—the frequency for infinitely small vibrations about the equilibrium point. For any real state specified by a positive integral v the oscillation frequency will be given by Eq. (3.14). Thus in the ground state ($v = 0$) we would have:

$$\bar{\omega}_0 = \bar{\omega}_e(1 - \tfrac{1}{2}x_e) \quad \text{cm}^{-1}$$

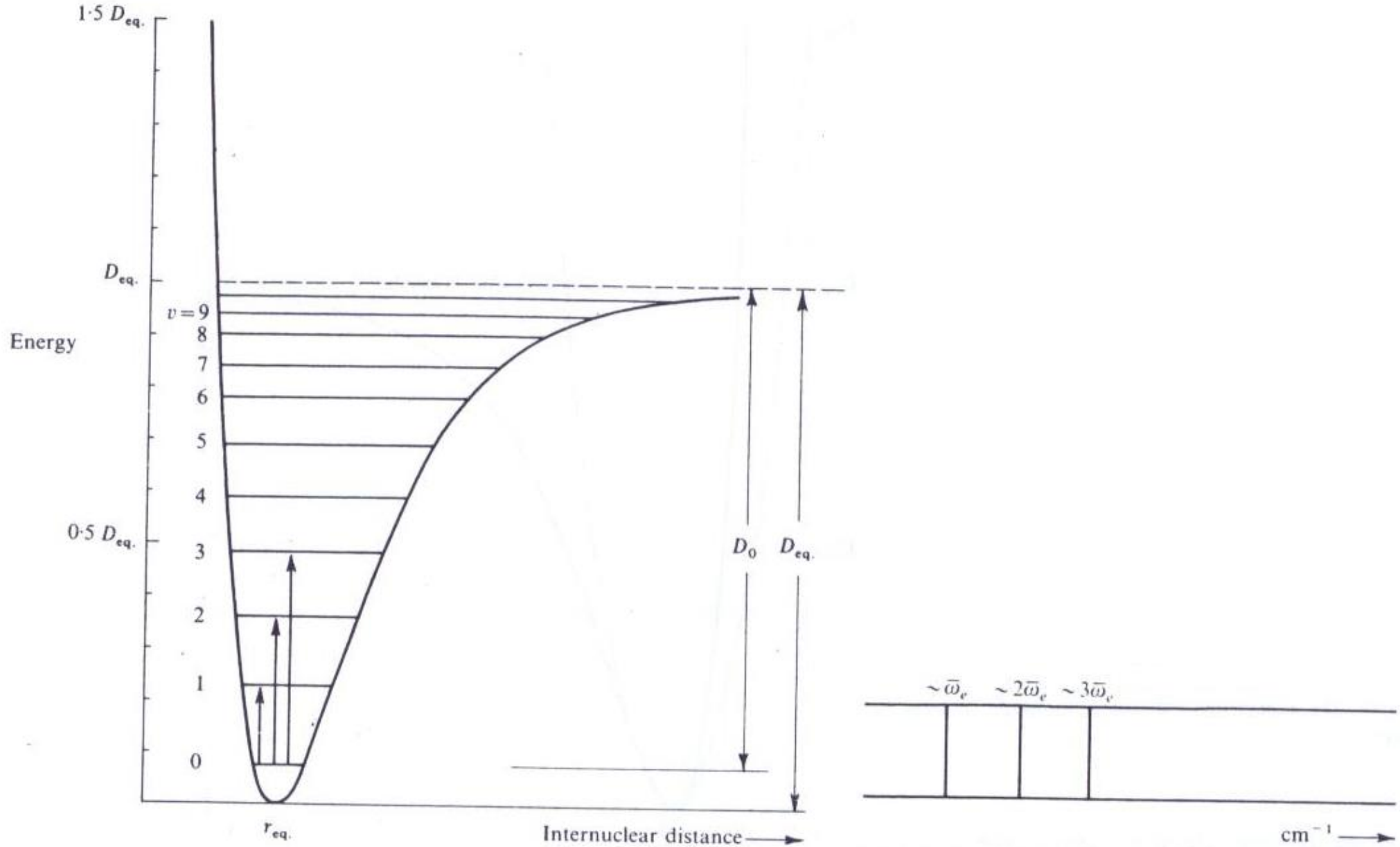


Figure 3.4 The allowed vibrational energy levels and some transitions between them for a diatomic molecule undergoing anharmonic oscillations.

and

$$\varepsilon_0 = \tfrac{1}{2}\bar{\omega}_e(1 - \tfrac{1}{2}x_e) \quad \text{cm}^{-1}$$

and we see that the zero point energy differs slightly from that for the harmonic oscillator (Eq. (3.7)).

The selection rules for the anharmonic oscillator are found to be:

$$\Delta v = \pm 1, \pm 2, \pm 3, \ldots$$

we may ignore all transitions originating at $v = 1$ or more and restrict ourselves to the three transitions:

1. $v = 0 \rightarrow v = 1$, $\Delta v = +1$, with considerable intensity.

$$\begin{aligned}\Delta\varepsilon &= \varepsilon_{v=1} - \varepsilon_{v=0} \\ &= (1 + \tfrac{1}{2})\bar{\omega}_e - x_e(1 + \tfrac{1}{2})^2\bar{\omega}_e - \{\tfrac{1}{2}\bar{\omega}_e - (\tfrac{1}{2})^2 x_e\bar{\omega}_e\} \\ &= \bar{\omega}_e(1 - 2x_e) \quad \text{cm}^{-1}\end{aligned} \tag{3.15a}$$

2. $v = 0 \rightarrow v = 2$, $\Delta v = +2$, with small intensity.

$$\begin{aligned}\Delta\varepsilon &= (2 + \tfrac{1}{2})\bar{\omega}_e - x_e(2 + \tfrac{1}{2})^2\bar{\omega}_e - \{\tfrac{1}{2}\bar{\omega}_e - (\tfrac{1}{2})^2 x_e\,\bar{\omega}_e\} \\ &= 2\bar{\omega}_e(1 - 3x_e) \quad \text{cm}^{-1}\end{aligned} \tag{3.15b}$$

3. $v = 0 \rightarrow v = 3$, $\Delta v = +3$, with normally negligible intensity.

$$\begin{aligned}\Delta\varepsilon &= (3 + \tfrac{1}{2})\bar{\omega}_e - \{\tfrac{1}{2}\bar{\omega}_e - (\tfrac{1}{2})^2 x_e\,\bar{\omega}_e\} \\ &= 3\bar{\omega}_e(1 - 4x_e) \quad \text{cm}^{-1}\end{aligned} \tag{3.15c}$$

These three transitions are shown in Fig. 3.4. To a good approximation, since $x_e \approx 0{\cdot}01$, the three spectral lines lie very close to $\bar{\omega}_e$, $2\bar{\omega}_e$, and $3\bar{\omega}_e$. The line near $\bar{\omega}_e$ is called the *fundamental absorption*, while those near $2\bar{\omega}_e$ *and* $3\bar{\omega}_e$ are called the *first* and *second overtones*, respectively. The spectrum of HCl, for instance, shows a very intense absorption at 2886 cm^{-1}, a weaker one at 5668 cm^{-1}, and a very weak one at 8347 cm^{-1}. If we wish to find the equilibrium frequency of the molecule from these data, we must solve any two of the three equations (cf. Eqs. (3.15)):

$$\bar{\omega}_e(1 - 2x_e) = 2886$$

$$2\bar{\omega}_e(1 - 3x_e) = 5668$$

$$3\bar{\omega}_e(1 - 4x_e) = 8347 \text{ cm}^{-1}$$

and we find $\bar{\omega}_e = 2990$ cm^{-1}, $x_e = 0{\cdot}0174$. Thus we see that, whereas for the ideal harmonic oscillator the spectral absorption occurred *exactly* at the classical vibration frequency, for real, anharmonic molecules the observed fundamental absorption frequency and the equilibrium frequency may differ considerably.

The force constant of the bond in HCl may be calculated directly from Eq. (2.22) by inserting the value of $\bar{\omega}_e$:

$$k = 4\pi^2\bar{\omega}_e^2c^2\mu \quad \text{N m}^{-1}$$

$$= 516 \text{ N m}^{-1}$$

when the fundamental constants and the reduced mass are inserted. These data, together with that for a few of the very many other diatomic molecules studied by infra-red techniques, are collected in Table 3.1.

Table 3.1 Some molecular data for diatomic molecules determined by infra-red spectroscopy

Molecule	Vibration (cm^{-1})	Anharmonicity constant, x_e	Force constant ($N\ m^{-1}$)	Internuclear distance $r_{eq.}$ (nm)
HF	4138·5	0·0218	966	0·0927
HCl†	2990·6	0·0174	516	0·1274
HBr	2649·7	0·0171	412	0·1414
HI	2309·5	0·0172	314	0·1609
CO	2169·7	0·0061	1902	0·1131
NO	1904·0	0·0073	1595	0·1151
ICl†	384·2	0·0038	238	0·2321

† Data refers to the ^{35}Cl isotope.

Although we have ignored transitions from $v = 1$ to higher states, we should note that, if the temperature is raised or if the vibration has a particularly low frequency, the population of the $v = 1$ state may become appreciable. Thus at, say, 600 K (i.e., about 300°C) $N_{v=1}/N_{v=0}$ becomes exp (−2·4) or about 0·09, and transitions from $v = 1$ to $v = 2$ will be some 10 per cent the intensity of those from $v = 0$ to $v = 1$. A similar increase in the excited state population would arise if the vibrational frequency were 500 cm^{-1} instead of 1000 cm^{-1}. We may calculate the wavenumber of this transition as:

4. $v = 1 \rightarrow v = 2$, $\Delta v = +1$, normally very weak,

$$\Delta\varepsilon = 2\tfrac{1}{2}\bar{\omega}_e - 6\tfrac{1}{4}x_e\bar{\omega}_e - \{1\tfrac{1}{2}\bar{\omega}_e - 2\tfrac{1}{4}x_e\bar{\omega}_e\}$$
$$= \bar{\omega}_e(1 - 4x_e) \quad cm^{-1} \qquad (3.15d)$$

We turn now to consider a diatomic molecule undergoing simultaneous vibration and rotation.

Roto-vibrational IR spectroscopy

In the IR absorption spectra recorded at **high resolution**, the vibrational bands have a structure of lines due to rotational transitions ($J_1 \rightarrow J_2$) that occur *simultaneously* with the vibrational transition ($v_1 \rightarrow v_2$).

At room temperature - only *the vibrational ground state* (v = 0) *is populated.*

- the rotational levels with J > 0 are populated: the rotational level with maximum population have a rotational quantum number different from zero ($J_{max} \neq 0$)!

$$E_{tot} = E_{vibr} + E_{rot}$$

IR radiation : $300 - 4.000\ cm^{-1}$

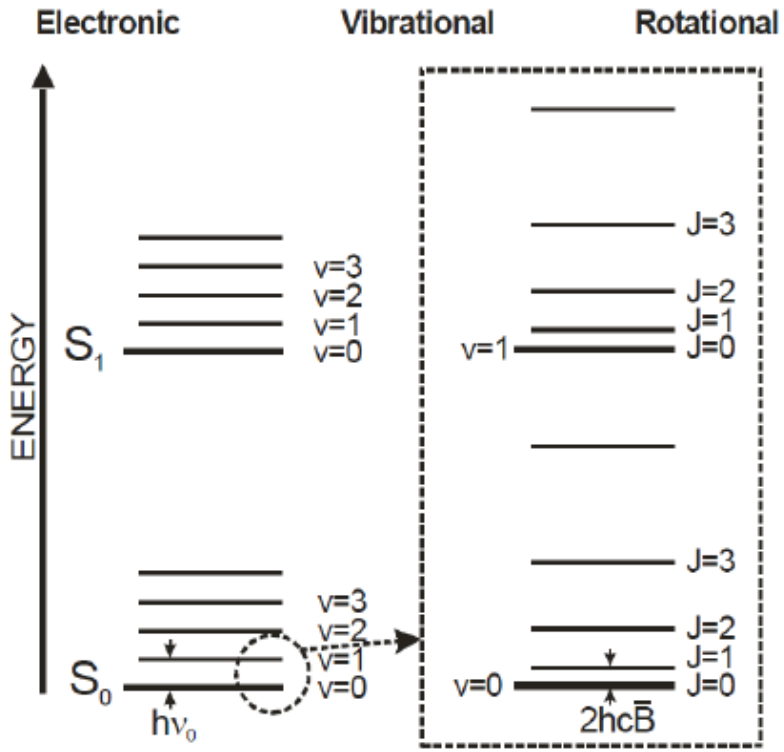
$$E_{tot} = h \cdot c \cdot \bar{\nu}_0 \left(v + \frac{1}{2} \right) + h \cdot c \cdot B \cdot J(J+1) + \ldots$$

$$\sim 1.000\,cm^{-1} \qquad \sim 1-10\,cm^{-1}$$

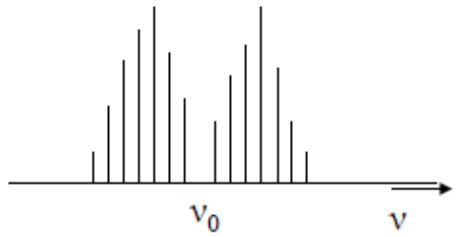
$$v = 0,1,2.... \qquad J = 0,1,2....$$

If the molecule (in vibrational ground state) *absorbed infrared radiation, it* will pass on an excited vibrational level ($\Delta v = +1$).

In the same time, the molecule can pass on another rotational level of excited vibration level (characterized by a *different rotational quantum number* than the initial) ($\Delta J = \pm 1$)



The spectrum consists of **lines** that appear at the **frequency** corresponding to **transitions**, having the *intensity proportional to the number of molecules that have made that transition*.



Pure vibrational spectrum: one line at ν_0.

Pure rotational spectrum: several lines separated by 2B.

Roto-vibrational spectrum : several lines separated by 2B, grouped into two branches (P and R), apart in ν_0 zone

(exception: for polyatomic molecule Q branch (ν_0) could appear!)

THE DIATOMIC VIBRATING-ROTATOR

Roto-vibrational spectra of diatomic molecules

Consider the molecule as an harmonic oscillator and a rigid rotor.

The selection rules for roto-vibrational transition are: $\Delta v = \pm 1 \qquad \Delta J = \pm 1$

The energy absorbed by molecule is equal with the sum of vibrational and rotational energy changes.

$$E_{abs} = \Delta E_{vib} + \Delta E_{rot} \qquad E(v,J) = h \cdot c \cdot [\overline{\upsilon}_0 \left(v + \frac{1}{2} \right) + B \cdot J(J+1)]$$

$$h \cdot c \cdot \overline{\upsilon} = E_2 - E_1 = E(v_2, J_2) - E(v_1, J_1) \rightarrow \text{absorbed energy}$$

$$\overline{\upsilon} = \frac{E_2 - E_1}{h \cdot c} = \left[\overline{\upsilon}_0 \left(v_2 + \frac{1}{2} \right) + B \cdot J_2(J_2+1) \right] - \left[\overline{\upsilon}_0 \left(v_1 + \frac{1}{2} \right) + B \cdot J_1(J_1+1) \right]$$

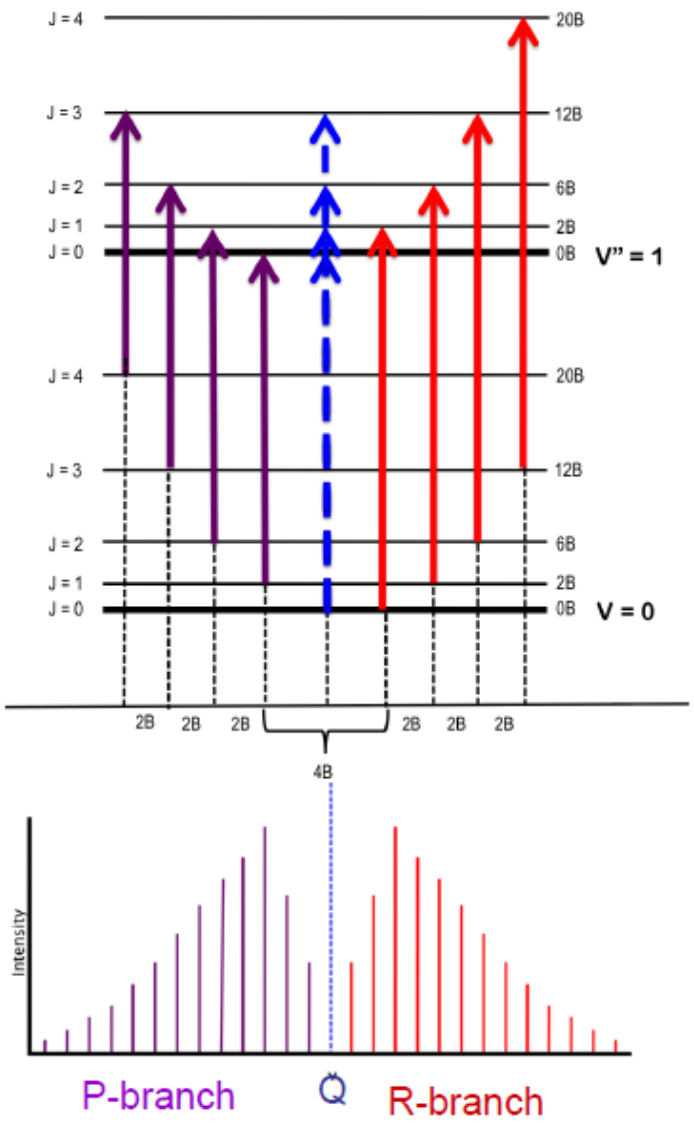
At room temperature: $v_1 = 0$ and **$\Delta v = +1$** so $v_2 = 1$

$$\overline{\upsilon} = \overline{\upsilon}_0 + B \cdot [J_2(J_2+1) - J_1(J_1+1)]$$ ← wavenumber of roto-vibrational transition

The rotational levels with $J \neq 0$ are populated, thus transitions with **$\Delta J = 1$** and **$\Delta J = -1$** may occur.

$\Delta J = +1 \rightarrow J_2 = J_1 + 1$ $\quad \overline{\upsilon}_R = \overline{\upsilon}_0 + 2 \cdot B \cdot (J_1 + 1)$ R branch ($J_1 = 0, 1, 2, \ldots$)

$\Delta J = -1 \rightarrow J_2 = J_1 - 1$ $\quad \overline{\upsilon}_P = \overline{\upsilon}_0 - 2 \cdot B \cdot J_1$ P branch ($J_1 = 1, 2, 3, \ldots$)



Selection rules: **ΔJ = 0, ±1**
Δv = 1

If ΔJ = +1 we obtained **R branch** ("rich", the molecule gain rotational energy)

If ΔJ = -1 we obtained **P branch** ("poor", the molecule lose rotational energy)

If ΔJ = 0 we obtained **Q branch** (for perpendicular vibrations in polyatomic molecules: *ex. NCN bend*)

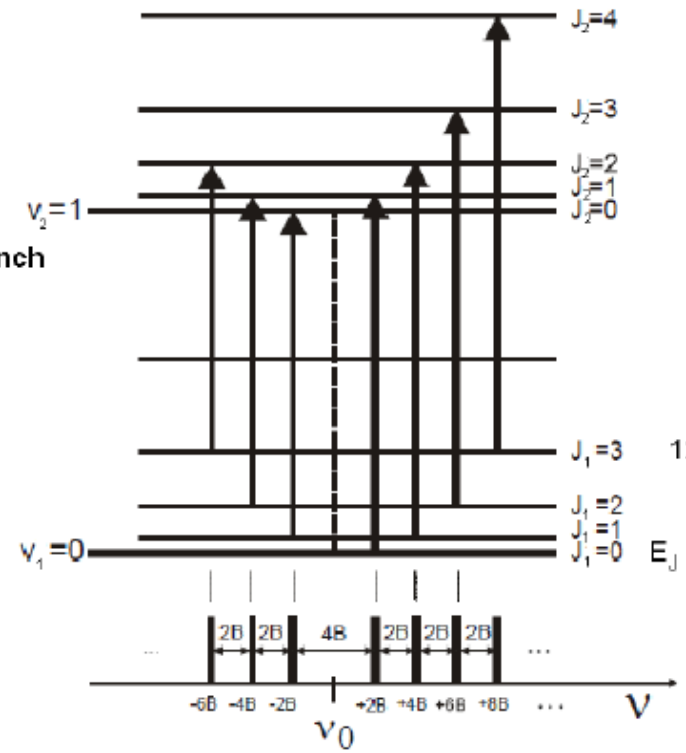
The lines of the P-branch (purple arrows) and R-branch (red arrows) are separated by 2B, thus the bond length can be deduced!

$$\overline{\upsilon}_{P,J} = \overline{\upsilon}_0 - 2 \cdot B \cdot J$$

J = 1, 2, 3, ...

$$\overline{\upsilon}_{R,J} = \overline{\upsilon}_0 + 2 \cdot B \cdot (J+1)$$

J = 0, 1, 2,



$$\overline{\upsilon}_{P,1} = \overline{\upsilon}_0 - 2 \cdot B$$

$$\overline{\upsilon}_{P,2} = \overline{\upsilon}_0 - 4 \cdot B$$

$$\overline{\upsilon}_{P,3} = \overline{\upsilon}_0 - 6 \cdot B$$

$$\overline{\upsilon}_{P,4} = \overline{\upsilon}_0 - 8 \cdot B$$

$$\overline{\upsilon}_{R,0} = \overline{\upsilon}_0 + 2 \cdot B$$

$$\overline{\upsilon}_{R,1} = \overline{\upsilon}_0 + 4 \cdot B$$

$$\overline{\upsilon}_{R,2} = \overline{\upsilon}_0 + 6 \cdot B$$

$$\overline{\upsilon}_{R,3} = \overline{\upsilon}_0 + 8 \cdot B$$

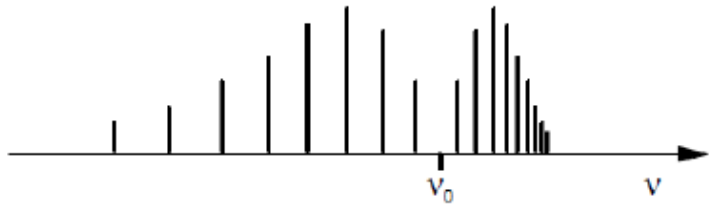
Q branch (ΔJ = 0) could appear in IR spectrum of polyatomic molecule!

Q tranzition (ΔJ = 0) is allowed only for vibrations that involved a perpendicular dipole moment change (to the symmetry axis of the molecule).

Q branch (ΔJ = 0): $\overline{\upsilon}_Q = \overline{\upsilon}_0$

For real molecule: vibration and rotation are not independent!

Two effects appear:



1) **roto-vibrational coupling**

Diatomic molecule: molecule vibrates → the bonds length (r) are changing → the moment of inertia changes → the rotational constant B changes

$$B_v = B_e - \alpha_e(v+\frac{1}{2})$$

α_e → roto-vibrational coupling constant
B_e → rotational constant for rigid and independent rotor

Each vibrational level have different rotational constant, so $B_0 \neq B_1$!

In this case the energy of roto-vibrational level is:

$$E(v,J) = h \cdot c \cdot [\overline{\upsilon}_0\left(v+\frac{1}{2}\right) + B_v \cdot J(J+1)]$$

$$E(v,J) = h \cdot c \cdot [\overline{\upsilon}_0\left(v+\frac{1}{2}\right) + B_e \cdot J(J+1) - \alpha_e(v+\frac{1}{2})J(J+1)]$$

2) centrifugal distorsion

When the rotation velocity increase (J increase) → the bond lenght increase → the moment of inertia increase → the rotational constant B decrease

$$B_J = B_e - D \cdot J(J+1)$$ D → centrifugal distorsion constant

In this case the energy of roto-vibrational level is:

$$E(v,J) = h \cdot c \cdot [\overline{\upsilon}_0\left(v+\frac{1}{2}\right) + B_J \cdot J(J+1)]$$

$$E(v,J) = h \cdot c \cdot [\overline{\upsilon}_0\left(v+\frac{1}{2}\right) + B_e \cdot J(J+1) - DJ^2(J+1)^2]$$

Considering both effects, the energy of a particular roto-vibrational level is:

$$E_{v,J} = h \cdot c\left[\overline{\upsilon}_0\left(v+\frac{1}{2}\right) + B_e J \cdot (J+1) - \alpha_e\left(v+\frac{1}{2}\right) J \cdot (J+1) - D[J(J+1)]^2\right]$$

$$E_{v,J} = h \cdot c\left[\bar{\upsilon}_0\left(v+\frac{1}{2}\right)+B_e J\cdot(J+1)-\alpha_e\left(v+\frac{1}{2}\right)J\cdot(J+1)-D\left[J(J+1)\right]^2\right]$$

We can consider: $E_{v,J} = hc\left[\bar{\nu}_0(v+1/2)+B_{v,J}J(J+1)\right]$

where: $B_{v,J} = B_e - \alpha(v+\frac{1}{2}) - DJ(J+1)$

P branch: ($\Delta v = 1,\ \Delta J = -1$) ($v = 0$ and $J = 1, 2, 3 \ldots$)

$$\bar{\nu}_{P,J} = \frac{1}{hc}(E_{(1,J-1)} - E_{(0,J)}) = \bar{\nu}_0 + (B_1 - B_0)J^2 - (B_1 + B_0)J$$

Q branch : ($\Delta v = 1, \Delta J = 0$) ($v = 0$ şi $J = 1, 2, 3 \ldots$) (if observable)

$$\bar{\nu}_{Q,J} = \frac{1}{hc}(E_{(v+1,J)} - E_{(v,J)}) = \bar{\nu}_0 + (B_1 - B_0)J(J+1)$$

R branch: ($\Delta v = 1, \Delta J = +1$) ($v = 0$ şi $J = 0, 1, 2, 3 \ldots$)

$$\bar{\nu}_{R,J} = \frac{1}{hc}(E_{(1,J+1)} - E_{(0,J)}) = \bar{\nu}_0 + (B_1 - B_0)(J+1)^2 + (B_1 + B_0)(J+1)$$

3.3 THE VIBRATION–ROTATION SPECTRUM OF CARBON MONOXIDE

In Fig. 3.7(*b*) we show the fundamental vibration–rotation band of carbon monoxide under high resolution, with some lines in the P and R branches numbered according to their J'' values. Table 3.2 gives the observed wavenumbers of the first five lines in each branch. We shall discuss shortly the slight decrease in separation between the rotational lines as the wavenumber increases; this decrease is apparent from the table and from a close inspection of the 'wings' of the spectrum.

From the table we see that the band centre is at about 2143 cm^{-1} while the average line separation near the centre is 3·83 cm^{-1}. This immediately gives:

$$2B = 3{\cdot}83 \text{ cm}^{-1} \qquad B = 1{\cdot}915 \text{ cm}^{-1}$$

Table 3.2 Part of the infra-red spectrum of carbon monoxide

Line	$\bar{\nu}$	Separation $\Delta\bar{\nu}$	Line	$\bar{\nu}$	Separation $\Delta\bar{\nu}$
$P_{(1)}$	2139·43		$R_{(0)}$	2147·08	
		3·88			3·78
$P_{(2)}$	2135·55		$R_{(1)}$	2150·86	
		3·92			3·73
$P_{(3)}$	2131·63		$R_{(2)}$	2154·59	
		3·95			3·72
$P_{(4)}$	2127·68		$R_{(3)}$	2158·31	
		3·98			3·66
$P_{(5)}$	2123·70		$R_{(4)}$	2161·97	

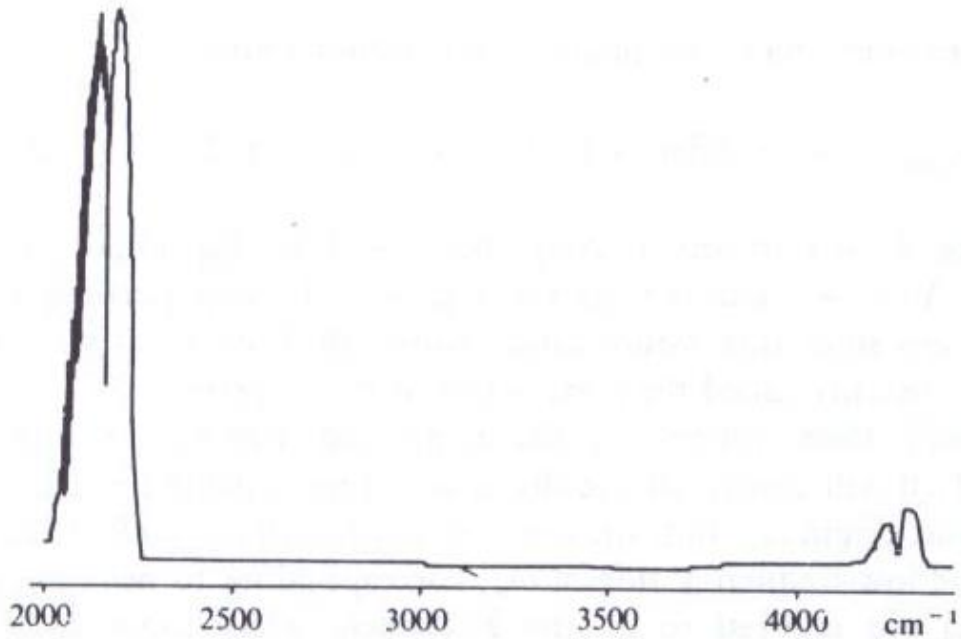


Figure 3.7(*a*) The fundamental absorption (centred at about 2143 cm^{-1}) and the first overtone (centred at about 4260 cm^{-1}) of carbon monoxide. The fine structure of the *P* branch in the fundamental is partially resolved. (Gas pressure 650 mmHg in a 10 cm cell.)

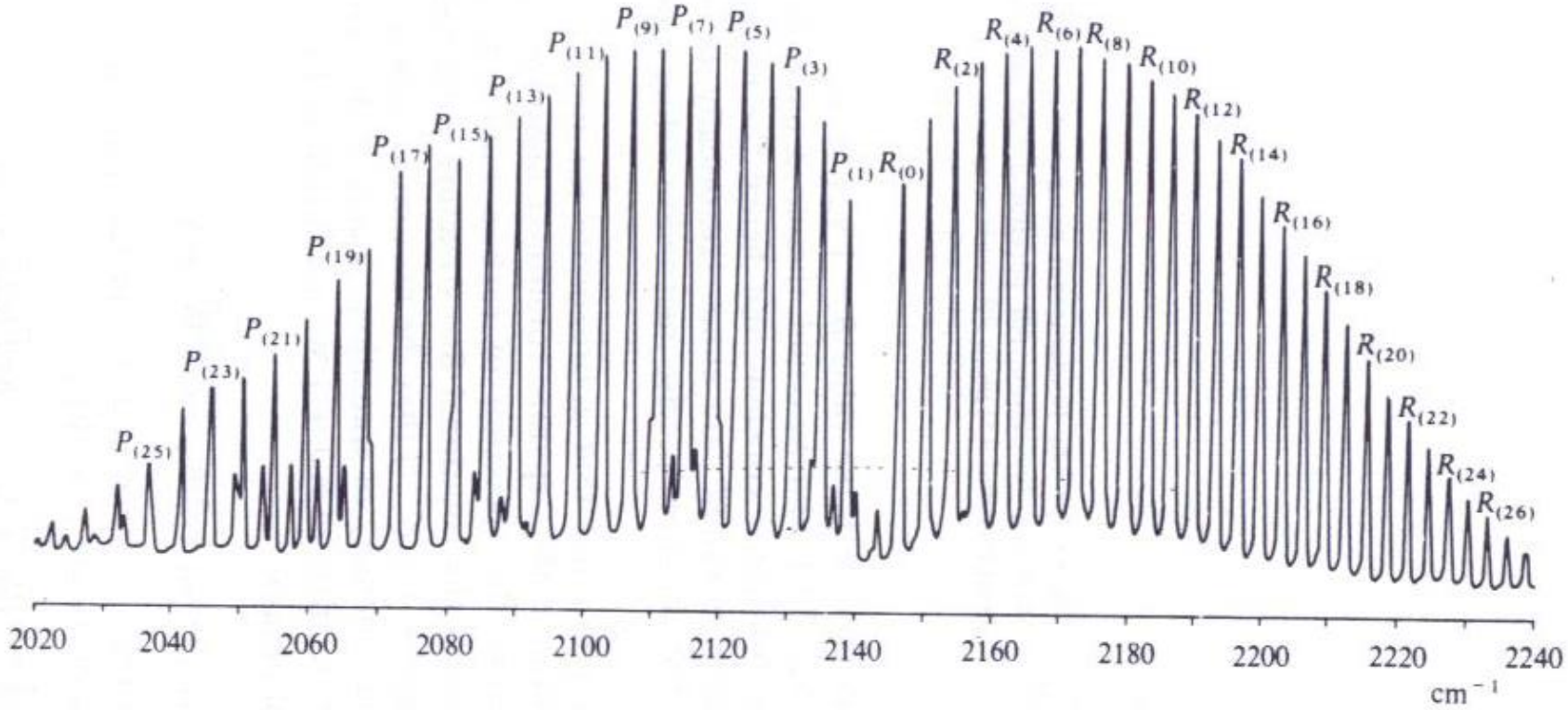


Figure 3.7(*b*) The centre of the fundamental band of carbon monoxide under higher resolution than in (*a*). (Gas pressure 100 mmHg in a 10 cm cell.) The lines are labelled according to their J'' values. The P branch is complicated by the presence of a band centred at about 2100 cm^{-1} due to the one per cent of ^{13}CO in the sample; some of the rotational lines from this band appear between P branch lines, others are overlapped by a P branch line and give it an enhanced intensity (e.g., lines $P_{(16)}$, $P_{(17)}$, $P_{(23)}$, and $P_{(24)}$).



Figure 3.8 The fundamental band of Fig. 3.7(*b*) under very low resolution. All rotational fine structure has been lost and a typical PR contour is seen.

This is in satisfactory agreement with the value $B = 1{\cdot}921\,18\ \text{cm}^{-1}$ derived by microwave studies (cf. Sec. 2.3.1) and we could, therefore, have obtained quite good values for the rotational constant and hence the moment of inertia and bond length from infra-red data alone. Historically, of course, the infra-red values came first, the more precise microwave values following much later.

It is worth noting at this point that approximate rotational data is obtainable from spectra even if the separate rotational lines are not resolved. Thus Fig. 3.8 shows the spectrum of carbon monoxide under much poorer resolution, when the rotational fine structure is blurred out to an envelope. Now we saw in Eq. (2.21) that the maximum population of levels, and hence maximum intensity of transition, occurs at a J value of $\sqrt{kT/2Bhc} - \frac{1}{2}$. Remembering that $m = J + 1$ we substitute in Eq. (3.20*c*) $m = \pm\sqrt{kT/2Bhc} + \frac{1}{2}$ and obtain:

$$\bar{\nu}_{\text{max. intensity}} = \bar{\omega}_0 \pm 2B(\sqrt{kT/2Bhc} + \tfrac{1}{2})$$

where the + and − signs refer to the R and P branches, respectively. The *separation* between the two maxima, $\Delta\bar{\nu}$, is then:

$$\Delta\bar{\nu} = 4B(\sqrt{kT/2Bhc} + \tfrac{1}{2}) = \sqrt{8kTB/hc} + 2B$$

or, since B is small compared with $\Delta\bar{\nu}$, we can write

$$\Delta\bar{\nu} \approx \sqrt{8kTB/hc}, \qquad B \approx hc(\Delta\bar{\nu})^2/8kT \quad \text{cm}^{-1}$$

where c is expressed in cm s^{-1}. In the case of carbon monoxide the separation is about 55 cm^{-1} (Fig. 3.8), while the temperature at which the spectrum was obtained was about 300 K. We are led, then, to a B value of about 1·8 cm^{-1} which is in fair agreement with the earlier values, but much less precise.

From Table 3.2 we see that the band origin, at the midpoint of $P_{(1)}$ and $R_{(0)}$, is at 2143·26 cm^{-1}. This, then, is the fundamental vibration frequency of carbon monoxide, if anharmonicity is ignored. The latter can be taken into account, however, since the first overtone is found to have its origin at 4260·04 cm^{-1}. We have:

$$\bar{\omega}_e(1 - 2x_e) = \bar{\omega}_o = 2143{\cdot}26$$

$$2\bar{\omega}_e(1 - 3x_e) = 4260{\cdot}04$$

from which $\omega_e = 2169{\cdot}74$ cm^{-1}, $x_e = 0{\cdot}0061$.

3.4 BREAKDOWN OF THE BORN–OPPENHEIMER APPROXIMATION: THE INTERACTION OF ROTATIONS AND VIBRATIONS

So far we have assumed that vibration and rotation can proceed quite independently of each other. A molecule vibrates some 10^3 times during the course of a single rotation, however, so it is evident that the bond length (and hence the moment of inertia and B constant) also changes continually during the rotation. If the vibration is simple harmonic the mean bond length will be the same as the equilibrium bond length and it will not vary with vibrational energy; this is seen in Fig. 3.1. However, the rotational constant B depends on $1/r^2$ and, as shown by an example in Sec. 2.3.4, the average value of this quantity is not the same as $1/r_{\text{eq.}}^2$ where $r_{\text{eq.}}$ is the equilibrium length. Further an increase in the vibrational energy is accompanied by an increase in the vibrational amplitude and hence the value of B will depend on the v quantum number.

In the case of anharmonic vibrations the situation is rather more complex. Now an increase in vibrational energy will lead to an increase in the average bond length—this is perhaps most evident from Fig. 3.4. The rotational constant then varies even more with vibrational energy.

In general, it is plain that, since $r_{\text{av.}}$ *increases* with the vibrational energy, B is *smaller* in the upper vibrational state than in the lower. In fact an equation of the form:

$$B_v = B_e - \alpha(v + \tfrac{1}{2}) \tag{3.22}$$

gives, to a high degree of approximation, the value of B_v, the rotational constant in vibrational level v, in terms of the equilibrium value B_e and α, a small positive constant for each molecule.

Here we restrict our discussion to the fundamental vibrational change, i.e., the change $v = 0 \rightarrow v = 1$, and we may take the respective B values as B_0 and B_1 with $B_0 > B_1$. For this transition:

$$\Delta\varepsilon = \varepsilon_{J',\,v=1} - \varepsilon_{J'',\,v=0}$$
$$= \bar{\omega}_o + B_1 J'(J'+1) - B_0 J''(J''+1) \text{ cm}^{-1}$$

where, as before, $\bar{\omega}_o = \bar{\omega}_e(1 - 2x_e)$.

We then have the two cases:

1. $$\Delta J = +1 \qquad J' = J'' + 1$$

$$\Delta\varepsilon = \bar{\nu}_R = \bar{\omega}_o + (B_1 + B_0)(J''+1) + (B_1 - B_0)(J''+1)^2 \text{ cm}^{-1}$$

$$(J'' = 0, 1, 2, \ldots) \quad (3.23a)$$

and

2. $$\Delta J = -1 \qquad J'' = J' + 1$$

$$\Delta\varepsilon = \bar{\nu}_P = \bar{\omega}_o - (B_1 + B_0)(J'+1) + (B_1 - B_0)(J'+1)^2 \text{ cm}^{-1}$$

$$(J' = 0, 1, 2, \ldots) \quad (3.23b)$$

where we have written $\bar{\nu}_P$ and $\bar{\nu}_R$ to represent the wavenumbers of the P and R branch lines respectively. These two equations can be combined into the expression:

$$\bar{\nu}_{P,R} = \bar{\omega}_o + (B_1 + B_0)m + (B_1 - B_0)m^2 \quad \text{cm}^{-1} \qquad (m = \pm 1, \pm 2, \ldots)$$

$$(3.23c)$$

where positive m values refer to the R branch and negative to P.

Table 3.3 Observed and calculated wavenumbers of some lines in the spectrum of carbon monoxide

m	J''	$\bar{\nu}_{obs.}$	$\bar{\nu}_{calc.}$†
30	29	2241·64	2241·91
25	24	2227·63	2227·65
20	19	2212·62	2212·54
15	14	2196·66	2196·53
10	9	2179·77	2179·66
5	4	2161·97	2161·90
0	—	(Band centre)	2143·28
−5	5	2123·70	2123·78
−10	10	2103·27	2103·40
−15	15	2082·01	2082·15
−20	20	2059·91	2060·02
−25	25	2037·03	2037·02
−30	30	2013·35	2013·14

† Values calculated from: $\bar{\nu} = 2143{\cdot}28 + 3{\cdot}813m - 0{\cdot}0175m^2$.

We see that ignoring vibration–rotation interaction involves setting $B_1 = B_0$, when Eq. (3.23*c*) immediately simplifies to (3.20*c*). Since $B_1 < B_0$ the last term of (3.23*c*) is always negative, irrespective of the sign of m, and the effect on the spectrum of a diatomic molecule is to crowd the rotational lines more closely together with increasing m on the R branch side, while the P branch lines become more widely spaced as (negative) m increases. Normally B_1 and B_0 differ only slightly and the effect is marked only for high m values. This is exactly the situation shown in the spectrum of carbon monoxide, Fig. 3.7(*b*).

In Table 3.3 some of the data for carbon monoxide are tabulated, together with the positions of lines calculated from the equation:

$$\bar{\nu}_{spect.} = 2143{\cdot}28 + 3{\cdot}813m - 0{\cdot}0175m^2 \quad \text{cm}^{-1}.$$

From this we see that, for this molecule:

$$B_1 = 1{\cdot}898 \text{ cm}^{-1} \qquad B_0 = 1{\cdot}915 \text{ cm}^{-1}$$

and hence, using Eq. (3.22), we have:

$$\alpha = 0{\cdot}018 \qquad B_e = 1{\cdot}924 \text{ cm}^{-1}$$

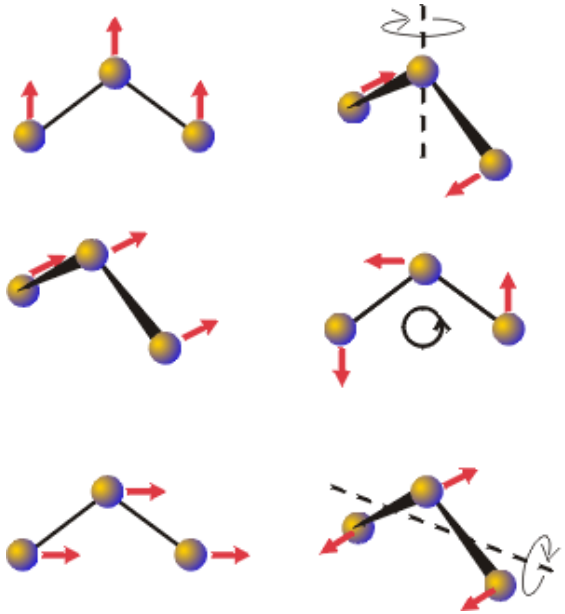
Further, we can calculate the equilibrium bond length and the bond lengths in the $v = 0$ and $v = 1$ states (cf. pp. 46–47) to be:

$$r_{eq.} = 0{\cdot}1130 \text{ nm} \qquad r_0 = 0{\cdot}1133 \text{ nm} \qquad r_1 = 0{\cdot}1136 \text{ nm}$$

The Vibration of Polyatomic Molecules

❖ How many vibrations are possible (=fundamental vibrations)?

For polyatomic molecules, the situation is more complicated because there are more possible types of motion. Each set of possible atomic motions is known as a *mode*. There are a total of 3*N* possible motions for a molecule containing *N* atoms because each atom can move in one of the three orthogonal directions (i.e. in the x, y, or z direction).



Translational modes

Rotational modes

A mode in which all the atoms are moving in the same direction is called a *translational mode* because it is equivalent to moving the molecule - there are three translational modes for any molecule.

A mode in which the atoms move to rotate (change the orientation) the molecule called a *rotational mode* - there are three rotational modes for any non-linear molecule and only two for linear molecules.

The other **3*N*-6** modes (or 3*N*-5 modes for a linear molecule) for a molecule correspond to vibrations that we might be able to observe experimentally. We must use symmetry to figure out what how many signals we expect to see and what atomic motions contribute to the particular *vibrational modes*.

Fundamental Vibrations

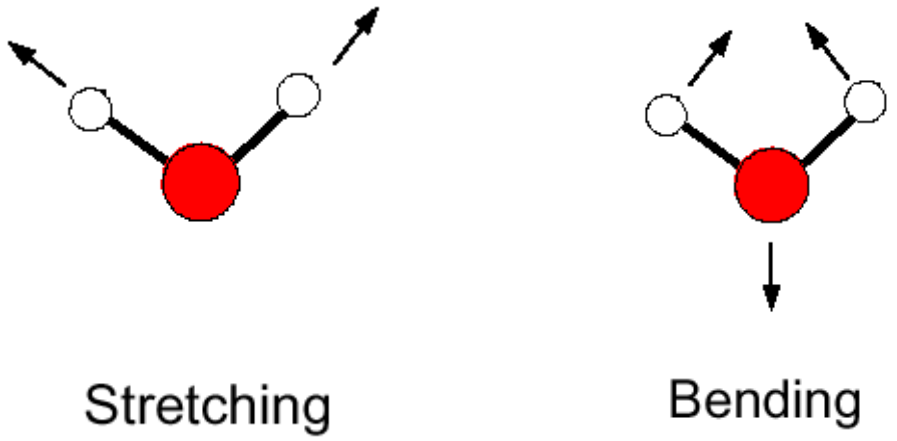
Degrees of freedom	linear	non-linear
Translational	3	3
Rotational	2	3
Vibrational	$3N–5$	$3N–6$
Total	$3N$	$3N$

N = number of atoms in molecule

Vibration Types

- There are two different types of vibrational modes:

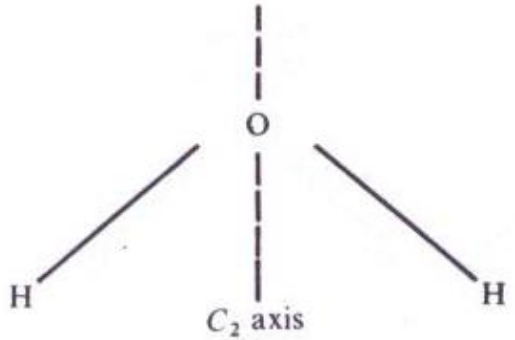
Vibrations can either involve a change in bond length (stretching) or bond angle (bending)

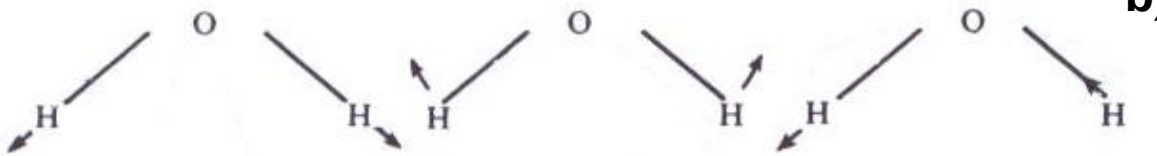


Stretching

Bending

The symmetry of the water molecule and its three fundamental vibrations

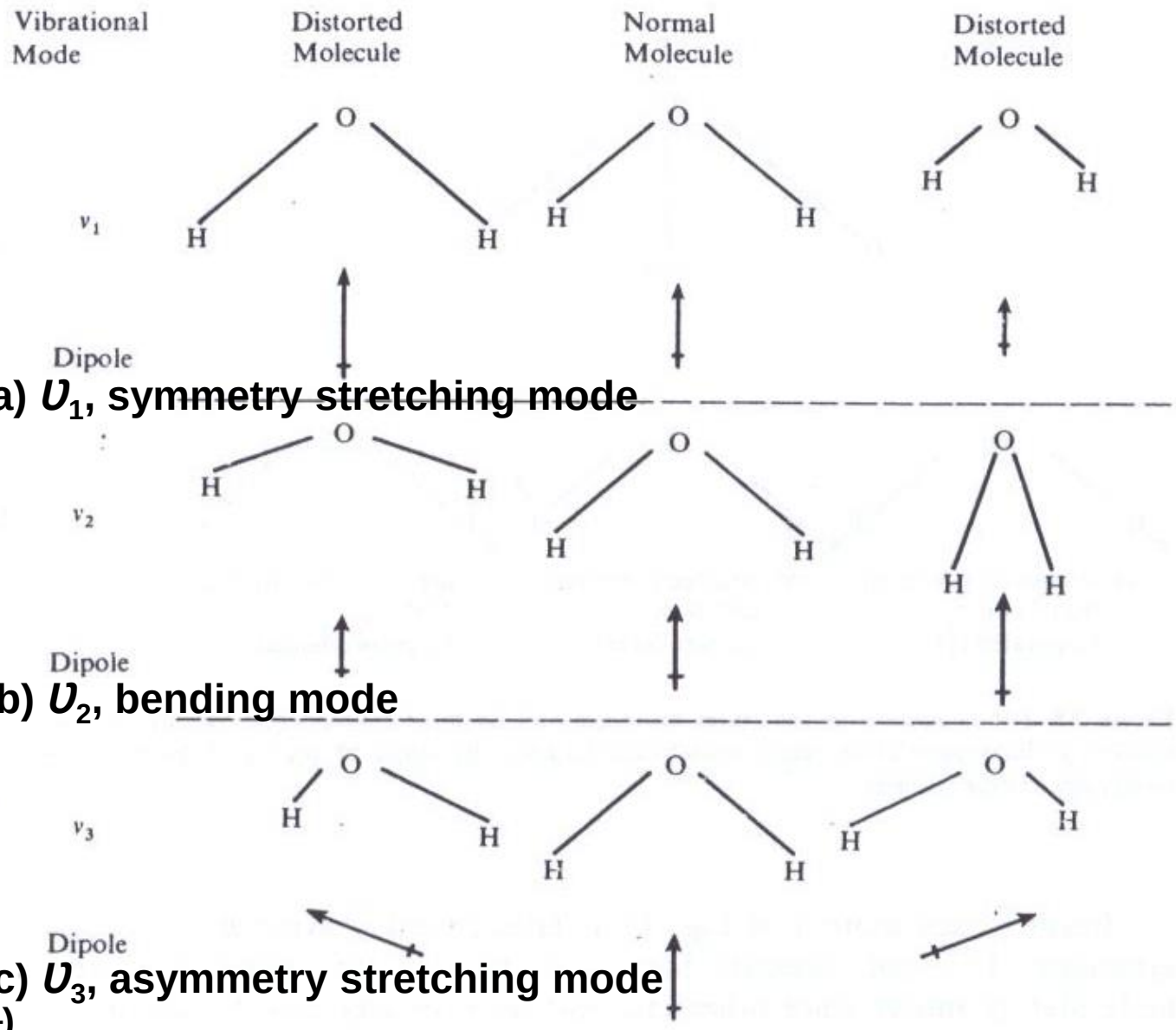




(a) symmetry stretch
3651.7 cm-1
υ_1, parallel (∥)

(b) symmetric bend
1595.0 cm-1
υ_2, parallel (∥)

(c) Antisymmetric stretch
3755.8 cm-1
υ_3, perpendicular(⊥)

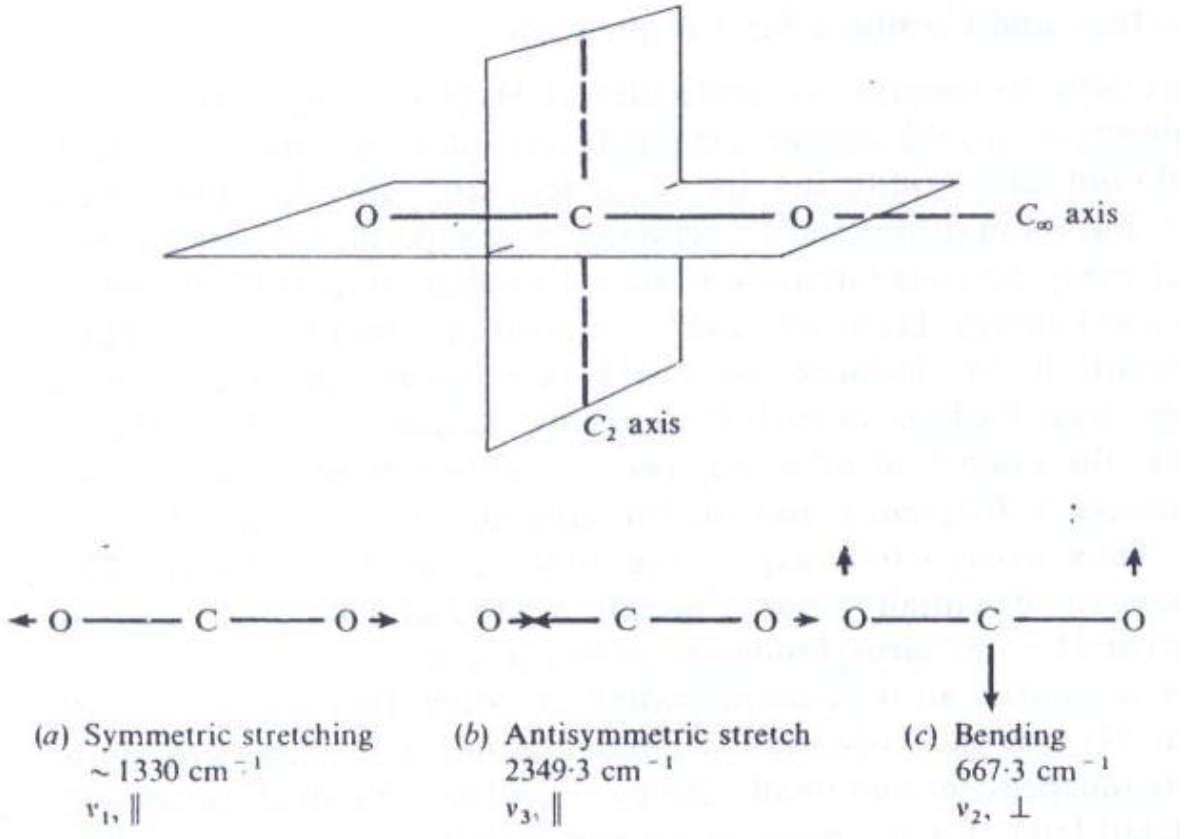


a) υ_1, symmetry stretching mode

b) υ_2, bending mode

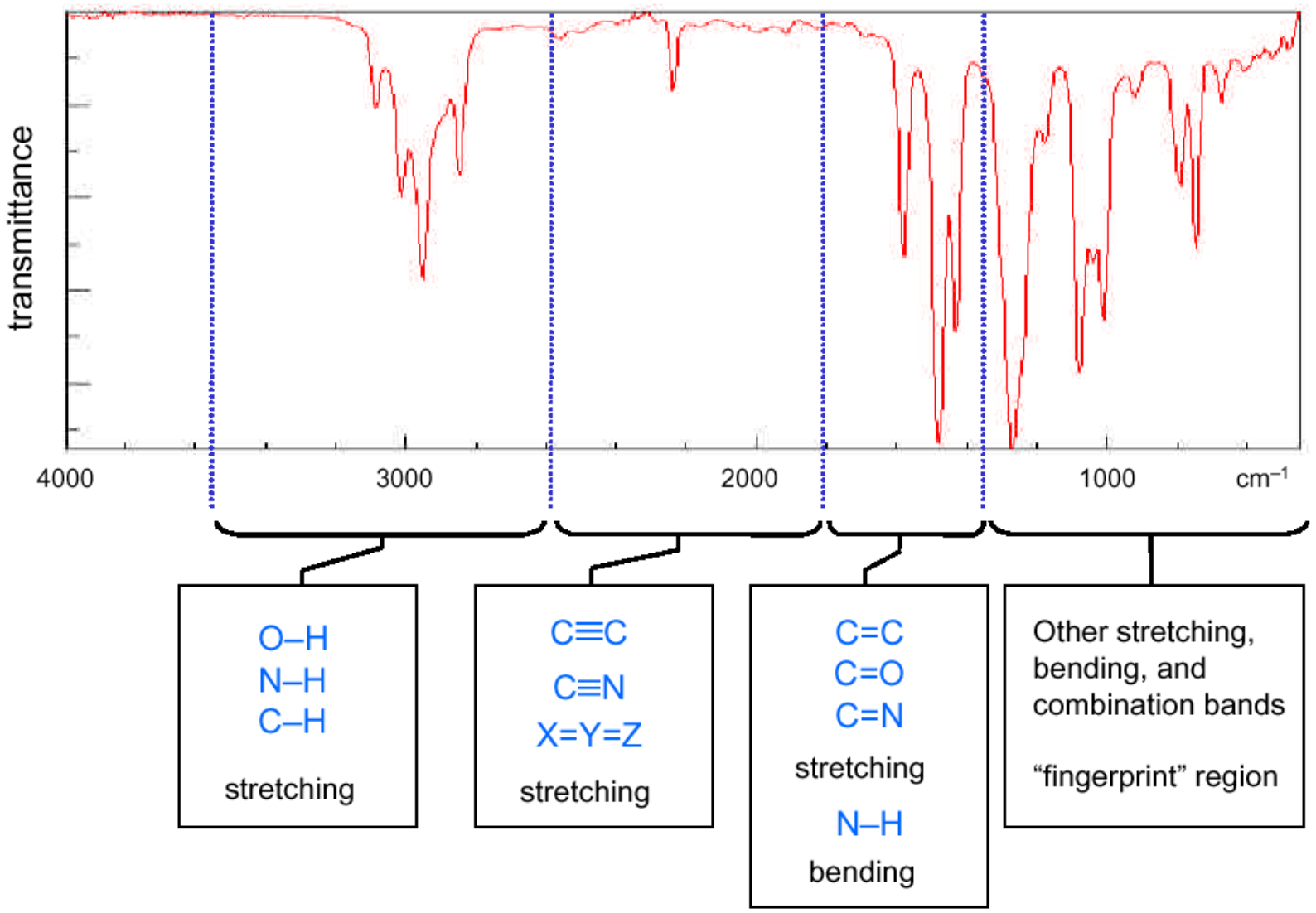
c) υ_3, asymmetry stretching mode

The change in the electric dipole moment produced by each vibration of the water molecule. This is seen to occur either (∥) or across (⊥) the symmetry axis.



The symmetry and fundamental vibrations of the carbon dioxide molecule.

Absorption Regions



Group frequencies

With certain functional or structural groups, it has been found that their vibrational frequencies are nearly independent of the rest of the molecule – group frequencies.

Carbonyl group 1650 to 1740 cm^{-1} various aldehydes and ketones

For many groups involving only two atoms, the approximate frequency of the fundamental vibration can be calculated from a simple *harmonic oscillator model*.

Calculations show that for most groups of interest, characteristic frequencies of stretching vibrations should lie in the region 4000 to 1000 cm^{-1}. In practical, the region from 4000 to 1300 cm^{-1} is often called the **group frequency region**.

The presence of various group vibrations in the IR spectrum is of great assistance in identifying the absorbing molecule.

TABLE 17-2 Abbreviated Table of Group Frequencies for Organic Groups

Bond	Type of Compound	Frequency Range, cm^{-1}	Intensity
C—H	Alkanes	2850–2970	Strong
		1340–1470	Strong
C—H	Alkenes 	3010–3095	Medium
		675–995	Strong
C—H	Alkynes (—C≡C—H)	3300	Strong
C—H	Aromatic rings	3010–3100	Medium
		690–900	Strong
O—H	Monomeric alcohols, phenols	3590–3650	Variable
	Hydrogen-bonded alcohols, phenols	3200–3600	Variable, sometimes broad
	Monomeric carboxylic acids	3500–3650	Medium
	Hydrogen-bonded carboxylic acids	2500–2700	Broad
N—H	Amines, amides	3300–3500	Medium
C=C	Alkenes	1610–1680	Variable
C=C	Aromatic rings	1500–1600	Variable
C≡C	Alkynes	2100–2260	Variable
C—N	Amines, amides	1180–1360	Strong
C≡N	Nitriles	2210–2280	Strong
C—O	Alcohols, ethers, carboxylic acids, esters	1050–1300	Strong
C=O	Aldehydes, ketones, carboxylic acids, esters	1690–1760	Strong
NO_2	Nitro compounds	1500–1570	Strong
		1300–1370	Strong

Fingerprint region

In the region from ≈ 1300 to 400 cm^{-1}, vibrational frequencies are affected by the entire molecule, as the broader ranges for group absorptions in the figure below – **fingerprint region**.

Absorption in this fingerprint region is characteristic of the molecule as a whole. This region finds widespread use for identification purpose by comparison with library spectra.

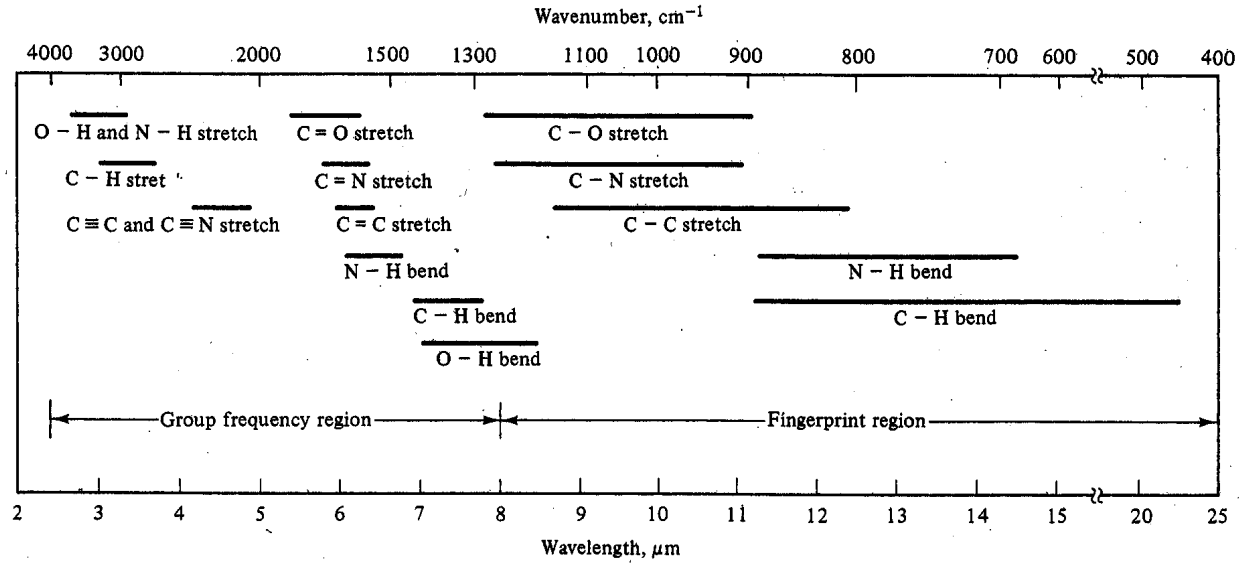
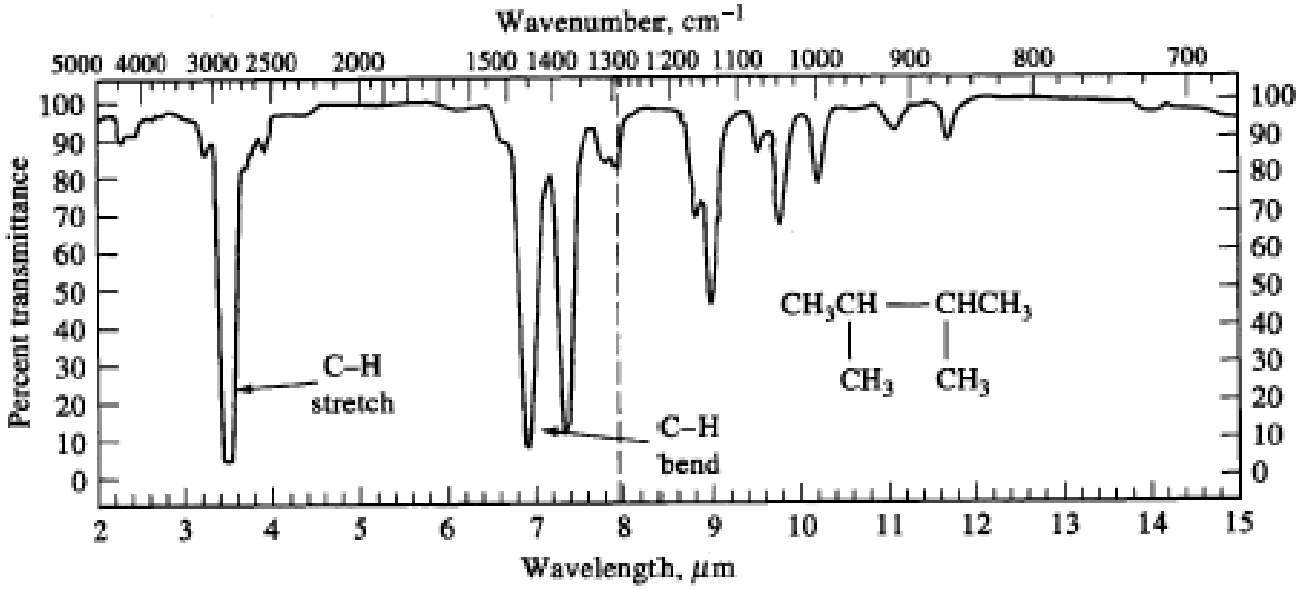
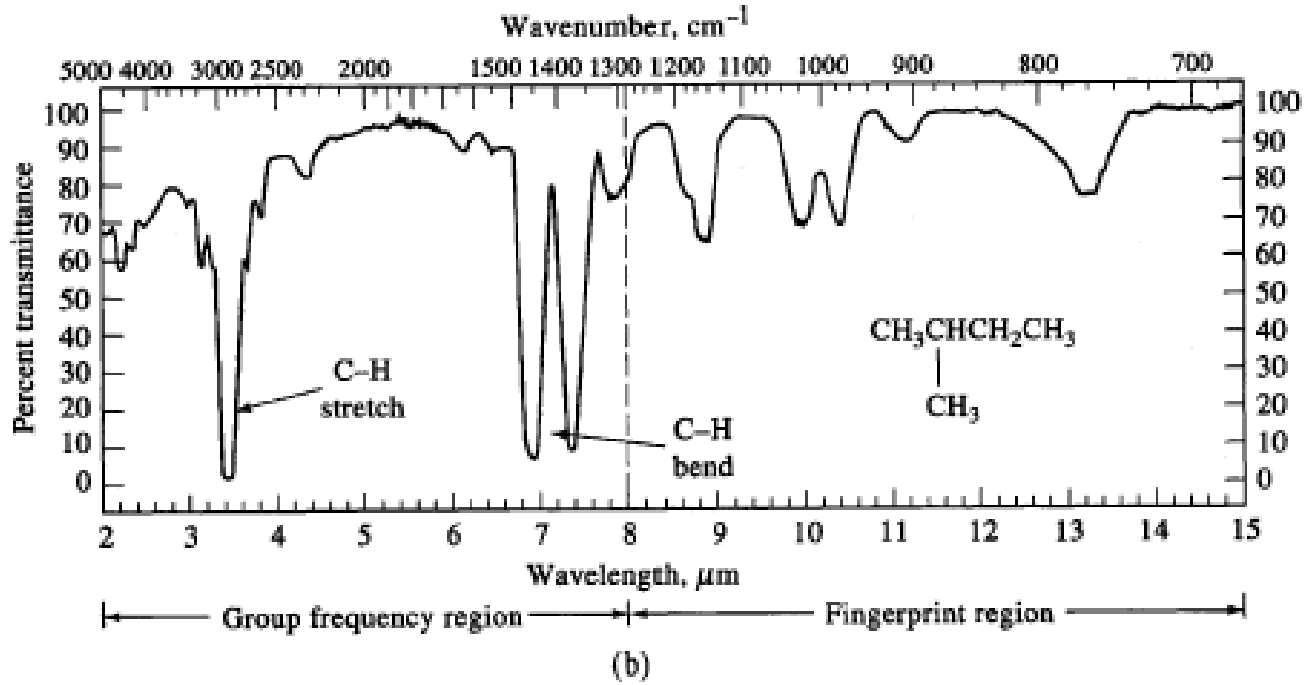


FIGURE 14-3 Frequencies of various group vibrations in the group frequency region and in the fingerprint region.


Wavenumber, cm⁻¹
5000 4000 3000 2500 2000 1500 1400 1300 1200 1100 1000 900 800 700
Percent transmittance
100 90 80 70 60 50 40 30 20 10 0
C–H stretch
C–H bend
CH3CH — CHCH3
CH3 CH3
2 3 4 5 6 7 8 9 10 11 12 13 14 15
Wavelength, μm

(a)


Wavenumber, cm⁻¹
5000 4000 3000 2500 2000 1500 1400 1300 1200 1100 1000 900 800 700
Percent transmittance
100 90 80 70 60 50 40 30 20 10 0
C–H stretch
C–H bend
CH3CHCH2CH3
CH3
2 3 4 5 6 7 8 9 10 11 12 13 14 15
Wavelength, μm
Group frequency region
Fingerprint region

(b)